



Smart Cultural Storyteller: *Bridging* Heritage and Technology

An AI-driven system for preserving and presenting cultural stories.

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Our Motivation: Preserving Stories in a Digital Age

The Essence of Heritage

Cultural and traditional narratives stand as the foundational pillars of our shared heritage, embodying invaluable lessons and identities from bygone eras.

Evolving Engagement

In the rapidly evolving digital landscape, modern audiences predominantly engage with rich, video-based content on platforms such as YouTube and Instagram.

The Risk of Obscurity

Regrettably, many traditional stories risk fading into obscurity or are not presented in compelling formats that resonate with today's digitally native audiences.

Labour-Intensive Creation

The conventional process of video creation—encompassing writing, recording, editing, and intricate design—demands considerable time and effort.

Our primary objective is to engineer an automated system that simplifies cultural storytelling, making it profoundly modern, engaging, and universally accessible.

Our Innovative Methodology: The Automated Pipeline

01

Input Handling

Users can provide either text or voice input. Voice commands are seamlessly converted to text using **Whisper STT**, whilst direct text prompts are processed immediately.

03

Visual Generation

Stunning illustrations are crafted using the **Stable-Diffusion-xl-base-1.0 model**. This process, demanding significant GPU resources, is efficiently executed on **Google Colab GPU**.

05

Narration & Audio

Natural, multi-language voiceovers are generated via **Edge TTS**, with **gTTS** serving as a reliable fallback. Background audio tracks are semantically selected to enhance the storytelling experience.

02

Story Generation

Our core text generation leverages the powerful **Falcon-H1-0.5B-Instruct** model. Enhanced with **Retrieval-Augmented Generation (RAG with FAISS)**, we minimise hallucinations and ensure profound cultural relevance.

04

Translation Layer

For consistent processing, all input is translated into English using **Google Translate**. The final output is then meticulously translated back into the user's chosen language, ensuring broad accessibility.

06

Video Assembly

The final video product seamlessly integrates narration, generated images, precise subtitles, and carefully selected background music, culminating in a professional, immersive experience.

Live Demonstration: Accessing the System

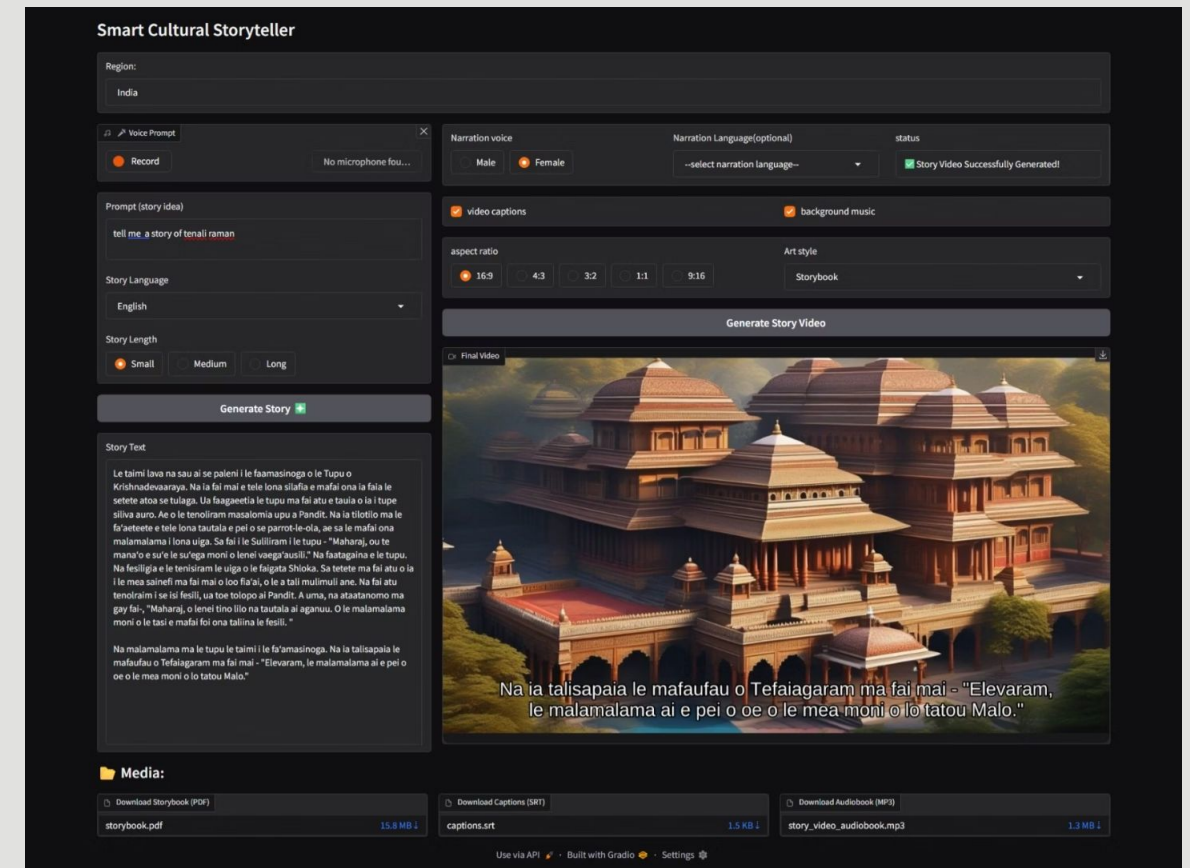
Our robust system has been successfully implemented and rigorously tested within the **Google Colab** environment.

The intuitive interface, meticulously built with **Gradio**, offers a dynamic and interactive user experience.

For a hands-on demonstration, the project notebook is readily available. You can access it directly via this link:

[Smart Cultural Storyteller – Colab Notebook](#)

Upon executing the notebook, a unique public URL is automatically generated, providing direct access to the Gradio interface. This link empowers users to effortlessly enter prompts, generate compelling stories, and create high-quality videos.



User Interface: The Intuitive Gradio Experience

The Gradio interface is designed for optimal user experience, segregating functionalities into distinct, easy-to-navigate panels.

Left Panel: Core Inputs

This section is dedicated to user input options, including the initial prompt, desired language for the story, the preferred story length,

Right Panel: Customisation Options

Here, users can fine-tune their output. Options include selecting the narration voice, choosing a distinct art style for visuals, defining the video aspect ratio, toggling captions, and selecting background music. and a real-time display of the processing status.

Integrated Video Preview

The final generated video is conveniently displayed directly within the same interface, allowing for immediate review and refinement before download.

Versatile Modes of Operation

The Smart Cultural Storyteller offers two distinct modes, catering to diverse creative workflows and user preferences.



Prompt-to-Story Mode

Users initiate the process by providing a brief idea or concise prompt. The system intelligently expands this into a fully developed narrative, which can then be reviewed and edited prior to video generation.



Paste-Story Mode

Alternatively, users can directly input a pre-written story, such as a traditional folk tale or a personal narrative. The system then seamlessly converts this existing text into a dynamic video, applying selected settings.

Comprehensive Output Features

Beyond the engaging video, our system provides a suite of versatile output formats, ensuring your stories can be enjoyed across various mediums and devices.



Full Storytelling Video

A complete video production, featuring rich narration, synchronised captions for accessibility, and harmoniously integrated background music, creating an immersive experience.



Audiobook (MP3)

Each story is also available as a high-quality MP3 audiobook, allowing listeners to enjoy the narrative on the go, perfect for podcasts or personal listening.



Storybook (PDF)

A printable PDF storybook, preserving the text and accompanying illustrations. Ideal for reading, sharing, or archiving for future reference and study.



Subtitle File (SRT)

A standalone SRT subtitle file is generated for each video, offering maximum compatibility and flexibility for re-editing or use with external media players.

Typical processing time for a complete story video ranges from **15 to 20 minutes**, depending on the narrative's length and complexity, ensuring efficiency without compromising quality.

Key Results and Observations

“The narration quality is remarkably natural, particularly excelling in its clarity and fluidity , providing an authentic listening experience.”

“Generated images consistently demonstrate an artistic flair and generally align well with the story's context, even if occasional minor discrepancies occur.”

“System performance, when evaluated against the capabilities of freely available tools, is notably satisfactory. Prospects for improvement are significant with more advanced models.”

Comprehensive demonstration outputs are readily available and can be reviewed within the dedicated project folder, showcasing the system's capabilities firsthand.

Conclusion: Pioneering the Future of Cultural Storytelling

The **Smart Cultural Storyteller** vividly demonstrates the transformative potential of Artificial Intelligence in safeguarding and promoting invaluable cultural heritage in contemporary, engaging formats.

By automating intricate processes, this system substantially **reduces the complexity** typically associated with creating storytelling videos, thereby making this powerful medium accessible to a much wider and more diverse audience.

Future Work

- Improve image consistency and narration quality.
- Enable background music and sound effects that change with scene mood.
- Fine-tune models specifically for cultural and regional folk stories.
- Expand multilingual support to additional regional and global languages.

Thank You!